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No. 39.—VOL. I.

SATURDAY, SEPTEMBER 27, 1884.

PRICE 1½d.

HOME-NURSING.

BY A LADY.

THIRD ARTICLE.

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If there are cupboards, they should be cleaned out before beginning to nurse, and in no case must they be used as receptacles for food or for anything that has become soiled in using. A chest of drawers is a comfort if used with discretion; but in home-nursing, it often proves the reverse of useful, and serves as a treasury for all sorts of things which have no business in a sick-room.

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much more of a difficulty than it need be; and though that relic of unenlightened days, the four-poster, is almost extinct in some households, most of the beds are large, and to alter the furnishing of a room seems beyond the imagination of the ordinary housekeeper. But surely, in time of illness, the first consideration should be how best to mitigate the patient's sufferings and lighten the nurse's labours; and in all but quite slight cases, it is worth considerable effort to secure the right kind of bed and bedding. This brings us to another home-nursing difficulty; for in spite of recent progress in hygienic knowledge, there are still not a few individuals who are foolish enough to prefer the unwholesome softness of a feather-bed to the healthy firmness of a good hair mattress. Yet few victims to this enervating habit fail to complain of the lumps, only to be got rid of by a large expenditure of strength and time. And if this be so in ordinary life, how much more complicated do matters become when the patient is too weak to bear to be removed for the necessary shaking, and is yet in a state to feel every discomfort with added force. Indeed, so true is this, that in serious illness, a feather-bed may be reckoned as an invincible enemy to comfort and cleanliness; and it is better not to allow your patient to remain under such unfavourable circumstances, even though it involve a few hours of dissatisfied grumbling. It may make things easier if the change is ushered in by allowing the feather-bed to remain under the mattress till the patient has become accustomed to the new order of things, when it will be well to substitute flock for feathers.

If the nurse's height will allow of two mattresses, it is a good plan to alternate them so that the under one of yesterday shall be the upper one of to-day. Those who have not tried this simple way of securing a change, will hardly believe the comfort it affords to a helpless patient.

On the same principle, it is a great alleviation to the monotony of long illness if the room will allow of two beds, one for day, the other for night; and this arrangement permits the thorough airing of bed and bedding, which it is otherwise so difficult to secure.

Bed-hangings are happily following the lead of the ancient four-poster, and will soon be only remembered as things of the past; and in illness, even the valance, still in use, had better be banished. Pillows should be firm and not too soft; and bolsters should have covers of their own, and not be rolled round in the under-sheet, according to a common and very shiftless process.

I have known the sheet to be tucked round the bolster in such a careless way that by the patient's merely getting into bed it has been pulled out of place, and has continued a fruitful source of irritation for the rest of the day.

How to keep the under-sheet smooth and free alike from crumbs and wrinkles is one of the professed nurse's arts, and is just one of those apparently small matters, neglected as such by inexperience, which make home-nursing the unsatisfactory thing it often is. Yet those who have gone through a tedious illness can hardly fail to remember the discomfort of what a tiny patient used to call 'rucks.' To him, poor little man, rucks made stopping in bed a misery; and his incessant demands to have them removed, suggested to even his inexperienced nurse the idea that there might be a right and a wrong way to such a commonplace thing as bed-making. The right and only proper way of arranging a bed for illness is to give special, minute attention to the placing of under-blanket and sheet. If the former can be dispensed with, especially in summer-time, it will be well; but very few patients will agree to this, and it is hardly of sufficient importance to argue about, except in special cases, when the doctor will probably give his veto.

In an ordinary way, be careful to choose a thin blanket, without holes of course, but one that has been pretty well worn, and that, therefore, will not greatly add to the heat of the patient's body. Then—more important still—the blanket must be considerably larger than the bed, in length as well as width. The mattress being arranged as evenly as possible, the under-blanket should be laid upon it, and well tucked in at top and bottom, as well as sides, till there is not a crease or wrinkle of any sort. The under-sheet must now be put on in the same way, and the nearer it resembles a drum in tightness the better. A few minutes spent in extra care at this point will save the patient hours of discomfort later on.

Should there, unfortunately, be no covered bolster, the sheet, after being securely tucked in at the bottom, must be drawn tightly upwards, and the bolster rolled round in the free end and then placed well up against the bed-head. The difficulty is to get the sheet tightly stretched without pulling the bolster away from its proper place. The number of pillows must be regulated by the nature of the patient's malady and partly by his previous habits. In ordinary cases, one large pillow is enough, and it is a good plan to place it lengthways, one end resting against the bed-head, and the other on the mattress, thus avoiding the hollow between the patient's shoulders and

the bed, which is a fruitful source of uneasiness and restlessness. In convalescence and in many forms of illness, pillows play an important part, and we shall have more to say about them later on.

We now return to the bed, furnished as far as the under-sheet and pillow. Should the patient be quite helpless, it will be necessary to place next, the contrivance known as a draw-sheet, which should consist of old linen or calico that has been washed to extreme softness. It should be folded lengthways till just long enough to reach from below the patient's shoulders to his knees. This, with an under-piece of mackintosh, is carefully put in place and tucked in with the same care and tightness as the under-sheet.

In regard to the upper bed-clothes, they too require care in tucking in, especially at the bottom; and if so long as to need folding back, the folding must on no account be done at the top. I have seen patients with chest complaints literally labouring under the weight of clothes placed in exactly the wrong place. In illness, as a rule the feet need extra covering, and the breathing organs no more weight of clothes than is absolutely necessary. The upper-sheet being light, may be allowed to turn over for neatness, but blanket and counterpane must only come up to the patient's neck. This folding back is such a common error, that it needs special notice, and in ordinary life, as well as in illness, should never be tolerated. The number of upper-clothes to be put on the patient's bed will vary with circumstances; but, as a rule, in a room with a fire, one blanket and a counterpane will be enough. The great point is to secure warmth without weight, and for this purpose, the ordinary cotton counterpane is about as bad as possible. An excellent substitute is the kind of blanket known as the 'Austrian,' which is cheerful in appearance as well as light and warm. In cases of exhaustion, nothing is better than a chintz-covered down, which can be shifted or removed without disturbing the patient.

A good many amateur nurses err on the side of over-heating a patient, and do not understand its connection with restless tossing and broken sleep. Should these symptoms occur at night, especially if accompanied with a flushed face and moist skin, it is well to gradually reduce the number of coverings, carefully noticing the effect of so doing.

On the ground of their greater warmth, cotton sheets and pillow-cases are preferable to linen; but some people have a decided fancy for the latter, and

it may safely be used, provided the sheets are warmed just before being put on the bed. Simple airing is not enough; I have known the chill of linen set a patient off into a shivering fit, although the nurse has been particularly strong in the airing line.

It is absolutely necessary that all the bedding used in illness shall be thoroughly aired, but, of course, it should never be done in the sick-room. Unhappily, it is by no means rare to find the fire screened from a patient by clothes-horse or chairs, covered with damp things, the vapour from which ought to be sufficient warning of the folly of such a practice. A good rule in this connection is that everything, down to cups and saucers, shall be removed from the sick-room as soon as soiled, and only returned to it in a condition for immediate use.

Re-making the patient's bed is our next consideration. If well enough to go into another room, he should either be carried there, or laid on a sofa and wheeled in. As soon as he is out of the way, the window of the sick-room should be thrown open to its widest, and the bed-clothes taken off one by one, well shaken, and left so exposed that the air can circulate freely around them. The mattresses should share the same treatment, and if possible, be left for a few minutes before being replaced. The patient will indeed be peculiar who does not enjoy the refreshment of a bed thus aired.

But in helpless illness, the changing of bedding is a more complicated matter, and needs practice to make perfect. There are two ways of changing the under-sheet. The first may be used when the patient is not quite helpless, and the nurse has to work alone. The soiled sheet is freed at the top, and after the removal of pillows and bolster, is rolled up to the patient's head; the clean sheet, after being well tucked in at the top, is loosely rolled in such a way as to lie close against the soiled one; they must now be worked down together, rolling the soiled, and unrolling the clean, the patient raising himself on elbows and feet just enough for the nurse to pass the sheets under him. In this way it is possible to get the under-sheet smooth and tight; but it is not an easy thing, and an assistant should be had if possible. If help is to be had, and in all cases where the patient is quite powerless, it is better to adopt the second plan. Remove the pillows and bolster, so that the patient lies quite flat in bed; turn him over on his side with his back to you. Loosen the sheet lengthways, and proceed with the rolling and unrolling as before, till the rolls come close up to the patient's back. Depress the mattress under

him, whilst the assistant draws the sheet through, and in so doing, slowly turns the patient on his back. He will now be lying on the clean sheet, and the difficulty is over. The trained nurse will be able to do this without removing the upper clothing, and in no case should all the coverings be taken away. Draw-sheets may be removed in the same way, but being small, are easier to manage. Some people tack or pin the clean to the dirty, and draw through whilst an assistant keeps the patient raised. Changing the upper clothes is not such a serious undertaking, though seldom properly managed by amateurs. The counterpane and blanket may be taken quite off the bed, and given to an assistant to shake, outside the room; but the sheet must never be removed without an immediate substitute. A good plan is to loosen the soiled sheet all round, tuck the clean one well in at the foot, and draw the free end upwards, under the dirty sheet, which is gradually drawn away or rolled up. As a general rule, the patient's bed should be made and his night-shirt changed at least once a day, and cases where this is not feasible ought not to come within the scope of home-nursing.

If a pair of clean sheets a day cannot be managed, one may be made to do, by letting yesterday's clean upper sheet be to-day's lower one; but draw-sheets must be changed as soon as soiled, irrespective of number. Where mackintosh is used, it should also be frequently changed, washed over, and thoroughly dried, in the open air if possible.

The patient's bed, it will be seen from the directions for making it, must never stand so that one side is against the wall, nor must it be in a direct current of air; but it is well if it can be so arranged as to face the fire and at the same time allow the patient to amuse himself by looking out of window. In badly finished houses, there is often considerable draught from cracks in door or window frame, and from this the patient must be carefully guarded by the judicious use of screens.

We now turn to consider how the sick-room may be kept in that state of perfect cleanliness essential alike to the patient's comfort and recovery; and of all neglected points, this is perhaps the one most frequently forgotten or ignored; not one in a hundred of home-nurses having a conception of her duty in this respect. Difficult it undoubtedly is; but where the patient can be removed to another room for an hour or two once a week, it is quite possible for even inexperience to be successful.

We will suppose the weekly removal has taken place, and the nurse has to make hay while the sun shines. She first strips the bed, sending the clothes into another room to be aired; and throwing the window open to its widest, she directs her attention to the grate. The best way of removing the ashes is to carefully collect all the large pieces of coal and cinder, and then very gently draw the ashes away into a piece of stiff paper, which folded over them, will prevent any dust rising in their transit. After cleaning grate and fire-irons and making up the fire, the nurse turns her attention to the carpet, which, after being well strewn with damp tea-leaves, should be briskly swept with a hand-broom. If the furniture is simple, it may be washed over with a wet cloth and dried, all cushions or stuffed furniture being beaten out of the room. The window, often overlooked, should be nicely cleaned; and then the bed being re-made, the patient may be brought back into a room thoroughly well cleaned and aired. It is not necessary that a nurse should herself perform all menial work; indeed, it is much better she should not; but she ought to see that the above directions are faithfully carried out. For the rest of the week, the carpet should be wiped over with a damp (not wet) cloth, tied to the end of a long broom, and the furniture well and quickly dusted. It is not enough to merely wipe over furniture and let the dust loose; the duster should be folded over bit by bit as it becomes soiled, and once or twice during the process of dusting, shaken out of a window in another room or in the staircase. A room thus treated will keep in perfect order for some time; but should the illness be long, an effort must be made to take the carpet up about every six weeks or two months, that it may be beaten and thoroughly aired. If carpets are made in the sensible fashion of squares, secured by brass nails with broad heads, there will be little difficulty in managing this; but it will not hurt the carpet to let it remain loose.

In cases where the weekly removal is impossible, the floor must be wiped over carefully every day with a damp cloth, and tea-leaves used now and then, the patient being protected from the dust by screens; but this plan is only for use as a last resource in extreme cases. Under such circumstances, it is not a bad plan to have a small portion of the furniture, say a chair or table and an ornament or two, removed each day and thoroughly cleaned, out of the room; otherwise, it is almost impossible to keep things in proper order, in spite of daily dusting.

BY MEAD AND STREAM.

CHAPTER XLVIII.—ANXIETIES.

MADGE wakened with the weary sensation of one who has passed through a long nightmare. It was some minutes before she could recall the incidents of the previous day; still longer before she could realise the unhappy meaning of the scene with Philip, and the fact that Uncle Dick and Aunt Hussy had found in her conduct cause for grave displeasure.

Surely she had been acting very wickedly, when those three, who were dearer to her than all the world beside, turned from her, and were vexed as well as pained by what she was doing, so far as they understood it. Surely Mr Beecham must be mistaken in the course he was pursuing—she did not even now doubt the goodness and generosity of his motives. There was only one way in which she could set the minds of her friends at ease; and that she must adopt, no matter what it might cost herself. She dare not hope that Philip would be readily satisfied and come back to her; but at least he should understand that she had been thinking of his interest more than of herself. And Uncle Dick and Aunt Hussy would be relieved from anxiety on her account, and then—who could tell?—maybe they would influence Philip. Maybe Uncle Dick would overlook his loss of fortune, and tell him that he never meant to separate them on such a sorry score as that.

The one way which she saw to bring about this desirable consummation was to inform Mr Beecham that she could no longer keep his secret; and that, if he did not come to Willowmere within the week to release her, she would take back the pledge she had given him and explain everything to her relatives.

Having arrived at this resolution, she was restored to a calm state of mind which was wonderful in contrast with the fever of the night. Morning is the time of hope and energy to a healthful nature; and Madge felt this, although the atmosphere was cold and the sky white with its load of snow, which was presently to descend in thick flakes, covering up the last patches of earth

and shrub left bare by the glimpses of sunshine that had succeeded the previous fall.

She went about her duties quietly and resolutely; but it was hard to meet the wistful eyes of Dame Crawshay without throwing herself into the arms that would have received her so gladly, and at once tell all. She had, however, made a bargain, and she would keep to it. Aunt Hussy would approve of what she was doing, when the time for explanation came. Uncle Dick was surly at breakfast, and he scarcely spoke to her at dinner. He did not once refer to the cattle show, and he went out to inspect his stock, a discontented and unhappy man.

Madge felt assured that Philip would say nothing more unless he heard from Uncle Dick; nevertheless, she was all day looking out for some sign from him. Old Zachy the postman came twice, and she saw him approach, her heart pausing, then beating quick with excitement. But Zachy brought nothing from her lover. And she was pained as well as disappointed, although she assured herself again that she had not expected anything, and that she had no right to expect anything until Philip received some token of Uncle Dick's kindly intentions. Besides, she argued, it was needful to bear in mind the distracted state he was in about his affairs, and how many things he must have to attend to which could not be postponed on any account. Indeed, she did remember all this, and was so keenly sensible of the cruel effect his misfortunes were producing on his mind, that she was frightened about him—more frightened than she had been even on the occurrence of the accident with the horse.

So, when postman Zachy had made his second and last round in the afternoon, she could not rest until she had consulted Dr Joy regarding Philip's health. Having explained to Aunt Hussy where she was going and why, she started for the village, although the snow had begun to fall in a way which would have made any town-miss who understood what the signs meant glad to stay at home. What the snow meant was to fulfil the threats it had been making for several days, and to come down more heavily than it had done for years.

Dr Joy was surprised to see her on such a gloomy afternoon; but he understood the nature of her visit, after a few words of such necessary explanation as she was at liberty to give.

‘And I want you to tell me plainly what his condition is, doctor,’ she concluded, ‘for I—that is, we are all very anxious about him.’

The good little doctor looked at her earnestly for a moment, as if to assure himself that she was not only desirous of hearing the truth but also able to bear it, and then made reply frankly, but was unable even then to dismount entirely from the hobby which he and his wife rode so diligently in theory.

‘You will agree with me, my dear Miss Heathcote, that economy is the great principle which should regulate our lives—not merely economy in finance, but likewise in work, in strength, and (most important of all) in health. I daresay our friend has told you that I spoke to him on this subject.’

‘When writing, he mentioned that you had visited him,’ she answered, with some nervous anticipations assailing her.

‘Well, I warned him then that his condition was extremely precarious. It is, in fact, that condition which when a man has fallen into, it requires him directly to throw up everything, if he cares to live. It requires him to sacrifice fortune, prosperity, and to run away anywhere and do anything to escape it.’

‘But how can he do that?’

Her own observations of Philip’s changing moods recently, formed a convincing argument in favour of the importance of what the doctor said. The doctor shook his head and smiled regretfully.

‘That is precisely what he asked; that is what every man to whom the advice is given asks. My answer is—*don’t* ask how, but go at once. Your affairs will be settled much more satisfactorily to all parties in a year or two if you go, than they can be if you remain and die in a month or two.’

‘But surely Philip is not so bad as that!’

‘You asked me to speak plainly, and I am quoting extreme cases,’ said Dr Joy, anxious to mitigate the alarm which he saw his verdict had created, whilst at the same time holding to his point. ‘Philip is not quite so bad as that yet; but he will be in a few months, unless something occurs to relieve him from his present anxieties.’

The doctor’s last words gave her more encouragement than he could have expected, or perhaps intended to give; and the terror which had made her

pulse seem to stop, was changed to confident hope. She had every reason to believe that in a few weeks, it might be in a few days, Philip would be relieved of all his anxieties. But this did not lessen in any degree her eagerness to have direct and frequent information as to the state of his health. Dr Joy readily agreed to call at the chambers in Gray's Inn on the following day, and report to her on his return; then they were to arrange about further visits. Thus being relieved to some extent on this important point, she prepared to take leave; but Dr and Mrs Joy suggested that she should have a fly to take her home, as the snow was falling fast, and already lay three or four inches deep on the ground, whilst it had drifted into an embankment against the opposite houses.

'I should not think of your hiring a conveyance,' said the doctor; 'but we have had a long and heavy day, and both my horses are fagged out.'

But Madge would not hear of this kindly proposal. 'I like the snow,' she answered, 'and a brisk walk will do me good.' At another time, she would have smiled at the timidity of her friends on account of the weather.

'You will catch your death of cold, my dear,' said Mrs Joy, 'and then you will not be able to come to Edwin's lecture next week. I assure you it is the most interesting one he has yet delivered.'

Even the danger of missing the doctor's lecture was not enough to deter her from walking home. As she was passing the *King's Head*, the Ringsford carriage drew up at the door, and out of it jumped Coutts Hadleigh, in the full uniform of a captain of Volunteers. He was taken by surprise, and uttered a natural exclamation:

'Why, what brings you so far from home on such an evening as this? There is going to be a regular out and outer of a snowstorm, and I would not be here myself, only this is the night of the feed I give every year to my men, and all the arrangements were made.'

She was more pleased to meet him than she was generally, for he might be able to give her some news of Philip. So, without troubling to answer his inquiries, she put her own.

'Don't know anything about him,' he answered—callously, as she thought, 'except that he has got into a precious scrape, and will disgrace our family, unless that uncle of his helps him out of it.'

‘Disgrace?—How is it disgrace to fail in a noble enterprise?’

‘Ah, it’s something worse than failing in a noble enterprise,’ answered Coutts, returning to his habitual tone of cynical indifference. ‘But don’t let us talk about it, if you please. I would rather not, even to you, until all the ins and outs are known.’

‘When will you know about your brother’s affairs?’

‘I cannot say; but he will tell you all about them; and if he doesn’t, I will. Meanwhile, let me do him a service—get into the carriage, and Toomey will drive you home. I am sure that is what Phil and the guv’nor, too, would say, if they found you trudging along the road in such weather. Do get in, or they will both have me down in their black books. The carriage is not to come back for me, so you won’t give the horses any extra work.’

She consented; and Toomey, who was glad enough to turn homeward for his own comfort as well as that of the horses, wheeled round, and drove off at a good pace. A little way out of the village they nearly ran over a man, who, walking in the same direction, had not heard the carriage making up on him, either on account of the preoccupation of his thoughts or the thick carpeting of snow on the road.

‘All right,’ growled the man, having saved himself, and Toomey drove on.

Madge recognised the voice of Caleb Kersey. She would have liked to speak to him, but it was too late. She supposed, however, that he was on his way to visit Sam Culver, from whom he would learn the cause of Pansy’s disappearance. Caleb was on this quest, as she surmised, and he was going to Ringsford, but not to seek information from the gardener.

CHAPTER XLIX.—AT MIDNIGHT.

Coutts Hadleigh relished good wine; but he was cautious in his cups, as in everything else. On this evening, however, he ‘drank fair,’ as it is called, with his comrades; and those who were acquainted with his habits noted the fact with increasing curiosity as the evening advanced. This was the fifth annual dinner he had given to ‘his men’ since the captain’s commission had been thrust upon him, and he had on no previous occasion displayed so much hilarity or provided so many cynical anecdotes for the entertainment

of the company. His lieutenant and sub.—both proprietors of the land they farmed—concluded that the captain must have made some exceptionally lucky stroke in business recently. Coutts believed he had.

The members of the Kingshope Volunteer corps were mostly young farmers and the sons of farmers, who should have possessed the physical proportions which would have specially qualified them for the soldier's career. But it was surprising to observe how few of them presented these qualifications. When Dick Crawshay first saw them mustered, he exclaimed in loud indignation, his huge form towering over the whole troop: 'What! is that all our county can show in the way of Volunteers? Why, half a dozen of our old yeomen would scare them into the middle of next week without a tussle! They are more like a set of town scarecrows than country-bred lads.... Ah, this comes ov givin' the land to people that have money and no muscle, and meddle with things they know nothing about.'

He was right in a certain degree, for these youths were the sons of wealthy merchants who take up farming as a hobby, and leaving the work to hired labourers, are indifferent to losses, and therefore able to pay rents which the working farmer has struggled for a time to compete with, and given up in despair, or emigrated. This was a sore subject with yeoman Dick, and although regularly invited by Coutts to this annual feast, he regularly refused to go—and even kept within his own bounds whenever he knew there was a parade. The prejudice prevented him from learning that a goodly number of these young fellows made up for physical deficiencies by skill as marksmen and efficiency in drill; so that the Kingshope Volunteer corps formed a by no means unsatisfactory body of men for home defence. But had any one dared to hint that even in some respects they might be favourably compared with the old yeomanry, he would have made Dick his foe.

On the present occasion, Captain Hadleigh's company showed that they had improved slightly on one of the yeomanry practices by keeping up their revels to a late hour without all getting drunk. The lieutenant having to pass Ringsford on his way home, and having his gig with him, drove the captain to the gates of the Manor. The snow had only ceased falling a little while before the company at the *King's Head* broke up, and now it lay deep on the roads, houses, and fields. The old church looked like a huge snow-house;

and the meadows in the dim moonlight presented a white surface, apparently on a level with the hedgerows.

The lieutenant's powerful cob had its work to do, for at every step its hoofs sank deep in the snow-covered road. But the travellers were merry, and did not mind the slowness of their progress. Their chief trouble was to keep the road and avoid the open ditches. They succeeded in this, and also succeeded in distinguishing the point where the Manor gates broke the white wall.

Coutts made his way through the side-gate, which shook large pancakes of snow down upon him as he opened it.

The avenue being guarded by its long arch of tree-branches, the path was comparatively easy to traverse, and Coutts was soon in front of the house, which, like the church, was a shapeless white mass, broken by a few points of light. Underneath these few lights was dark shadow. As Coutts ascended the steps of the portico, a man stepped out from the shadow.

'I want to speak to you a minute, Mr Coutts Hadleigh; I've been waiting all evening for you.'

Coutts was no coward, although his brain was somewhat muddy with wine; but this sudden apparition made him spring to the top of the steps and ring the bell, as he exclaimed fiercely:

'Who are you, and what do you want with me at this hour?'

'I want to know where is Pansy Culver?' said the man with enforced calmness, which contrasted to his advantage with the blustering ire of the other.

'Confound your impudence—how should I know?'

'I saw you with her at the London station. Where has she gone to? Where did you send her to?'

'She didn't tell me where she was going to, and I didn't send her anywhere.'

Caleb Kersey's calmness broke bounds, and he next spoke with savage determination:

'You are lying, and you shall tell me the truth.'

‘You’re an insolent fool.’

As Caleb swiftly ascended the steps, he received a vigorous buffet on the breast, which tumbled him backward on the snow. The door was open; Coutts entered; the door was instantly closed, bolted, chained, and locked.

‘Tell that fellow Kersey to go about his business,’ said Coutts to the attendant who had been waiting up for him; ‘he is drunk or mad. If he has any business with me, he knows where to find me at proper hours.’

With that he went up-stairs in a furious temper with the man who had insulted him, and had evidently intended to offer violence to his person. Before he had reached the first landing, there was an impatient but not a very loud knock at the door. The servant repeated his young master’s message, put out the hall lights, and gladly enough went off to bed.

Caleb stood in the portico hesitating as to what he should do. He had been waiting there for hours; he had been told that Mr Coutts Hadleigh was not at home—the servant declined to say where he might be found. The snow and the cold did not appear to affect him. He waited, and at last the man had come, but had not given the watcher any satisfaction. Caleb was aware that his application was untimely: but that was not his fault: the circumstances were exceptional. He must know from this villain what he had done with Pansy, and then he would seek her father, whose authority would rescue her from the evil influence under which she had fallen.

The poor fellow never thought that his first step ought to have been to consult Pansy’s father. A natural delicacy, rude and earnest, made him shrink from the idea, because he felt sure it would cause him pain. He learned from his friends in the village that Pansy had gone away somewhere; and as the gardener had no special need or liking to speak of her grandfather, he had not mentioned to any of his gossips whither she had gone. So Caleb, sitting in a train which was just starting, having caught sight of Pansy and Coutts Hadleigh talking together on the platform at Liverpool Street Station, instantly concluded that there was something wrong. He would have jumped out of the carriage; but the other passengers prevented him, and he had to endure cruel torments of speculation and rage until he reached his destination.

He had no hope of winning Pansy; but he might save her from the fate to which she seemed to be hasting. He had no doubt she had been taught to repeat some falsehood to her father, which kept him quiet about her absence, and he had no doubt of her danger. Then with a sullen resolution, in which the anxiety of a lover was combined with the suppressed fury of a maniac, he sought Coutts Hadleigh, determined to force the truth from him.

In those cold weary hours when he was hanging about the Manor waiting, the words of Philip frequently recurred to him: 'Trust her, man; trust her.' He imagined that he did trust her; he was sure that she did not mean to do wrong. But at the same time the wicked comment of Wrentham also presented itself, reminding him that trust gave the woman opportunity to deceive. He did not like the man who spoke or the words he uttered; but the remembrance made him uneasy.

'Ah, if Master Philip had not been in such a pickle with his own affairs, I'd have gone to him now, and he would have told me what was best to do, even though the villain be his own brother. But it would be a mortal shame to put more trouble on him when he's down enough already. I'll go my own way.'

All these things were careering through his mind, as he stood under the portico wondering how he should act. He heard a casement open above—it seemed to be directly over his head—and Captain Hadleigh shouted:

'You'd better move off quietly, Kersey, or I'll call our fellows and send for the police.'

The casement was closed violently, the two sides banging together, the principal windows of the Manor opening on hinges like doors, in the French fashion.

Caleb stepped out from beneath the portico and looked up. There was a ruddy glow—the effect of the light shining through deep maroon-coloured curtains—in two windows on the first floor. One of these windows opened on to the top of the portico which formed part of a balcony. That was the one from which Hadleigh must have spoken, thought Caleb; and was immediately satisfied on the point by seeing the shadow of a man who was passing slowly between the light and the curtains.

‘The stable ain’t far off, and I can find a ladder there,’ muttered Caleb, moving away from the front of the house.

Mr Hadleigh, sen., was seated at his writing-table, his back towards the windows. Before him lay those sheets of manuscript which he had written at intervals during the past year. The broad shade on the lamp cast the soft light down on the table, and had it not been for the bright glow of a huge fire, the rest of the room—and especially the upper part—would have been in comparative darkness. As it was, the flickering flame of the fire made the shadows above and around him flutter and change like living things.

He was not writing. He was carefully separating certain pages from the others; having done so, he fastened them together neatly, and with his hand covering them, as if to hide the words from himself, he leaned back on his chair. Suddenly he rose and paced the floor slowly in melancholy reflection.

When he resumed his seat, there was a placid expression on his face, like that of one who, after a long mental struggle, has come to a final decision and found peace.

With as much sad deliberation as if he were committing a dear one to the grave, he placed the separate packets of manuscript in different envelopes. The first and largest he addressed in a bold clear hand, ‘*To Mrs Philip Hadleigh. To be opened after my death.*’

Over the second packet his pen was poised for some moments, and his hand was not so steady as before when he began to write.

‘*To my son, Philip Hadleigh. To be opened after my death and read by him alone. When he has read, he shall decide whether to burn at once or first to show it to his wife. The secret of my life is here.*’

As his pen stopped, a chill blast passed through the room, making the lamplight waver, as if it were about to be extinguished. Mr Hadleigh, surprised, raised his head slowly, and slowly looked round.

The window behind him was open, and before it stood a tall, rough-looking, muscular man. Mr Hadleigh’s sallow cheeks became more sallow, his eyes

started, and his lips trembled slightly. He recovered himself instantly, and rising calmly from his seat, and at the same moment lifting the shade from the lamp, his eyes remaining fixed all the time on the intruder, burglar, intending murderer, perhaps.

When the light was uncovered, the man drew back a pace with a kind of growl of surprise. Mr Hadleigh retained perfect self-possession; but he was not much relieved from apprehension by recognising in his midnight visitor the leader of the agricultural agitators who had on various occasions openly declared antagonism to the master of Ringsford.

(To be continued.)

A LITTLE KNOWLEDGE *NOT* DANGEROUS.

A REMARKABLE circumstance recently occurred which brings out strongly the fact that scientific teaching in medical and surgical matters has made giant strides of late. On the 8th of July an accident happened to a 'marker' at the ranges of the Civil Service Rifle regiment at Wimbledon when marking at the five hundred yards' range. According to the report, a rifle-bullet seems 'to have bounded off the corner of the target and to have entered the marker's breast.' Fortunately, the great annual meeting of the National Rifle Association was to commence in a day or two, and the Field Hospital prepared for the meeting was being got ready under the charge of Sergeant Monaghan and Corporal Melville, both of the Army Hospital Corps. Thither the wounded man was immediately carried; but there was no surgeon present or anywhere near. Seeing, however, the serious nature of the case, the two soldiers, without a moment's hesitation, took steps to extract the bullet, which had entered the right breast just under the collar-bone. Having carefully examined his patient and found the exact locality of the bullet, Sergeant Monaghan, with the assistance of the corporal, made an incision in the back and was enabled at once to extract the bullet from the spot where it had lodged, just opposite to the point of entry in the breast. The injured man, a member of the corps of Commissionaires, expressed himself much gratified with the prompt attention he had received, as well as with the skilful operation by which, without a moment's loss of time, the important act of removing the bullet had been accomplished. Too much praise cannot be given to the two soldiers, who by their ready and intelligent action, saved their patient not only from prolonged suffering, but perhaps even from death itself.

The well-known saying of 'A little knowledge is a dangerous thing,' is here singularly confuted, for it was just the 'little knowledge' applied with sagacity and intelligence that probably saved the life of a fellow-creature.

Many of our readers will doubtless remember a melancholy occurrence which took place last year on one of the Swiss mountains, when a valuable young life was lost for the want of a 'little knowledge,' and in itself, very

simple knowledge too. A German engineer and two guides were ascending one of the famous Swiss mountains, when the younger of the guides appears to have had a very bad fall, by which either a bottle or a lamp-glass was broken, the fractured part entering the young man's thigh and dividing the femoral artery. It would not, we should suppose, have required very profound surgical knowledge to know that the man would inevitably bleed to death unless this great artery could be immediately compressed; but incredible as it may appear, neither the German nor the other Swiss guide knew anything about the matter. They tried to stop the spouting blood with their handkerchiefs, which of course was of no avail. Neither thought of tying the handkerchief or other ligature round the upper part of the limb, and then twisting it tight by the application of a stick; and so the poor young fellow quickly bled to death. Now, if the bleeding could have been arrested by ligature until surgical assistance was procured, the young guide would doubtless have recovered, for the injury, as a mere flesh-wound, was in itself by no means serious. Here, then, a 'little knowledge' would have done a vast amount of good.

One of the best, most useful, and practical associations of the present day is the St John's Ambulance Society, which teaches all who care to learn how to act in such emergencies as that related, and to take instant action *on the spot*, until surgical aid can be obtained—a ticklish and anxious time, often fraught with serious danger, when there is not a minute to spare, and where loss of time means loss of life.

Let every one, therefore, who has any real love for his fellows, and who feels that he or she has the nerve requisite for the work—for this is a *sine qua non*—at once learn how to act in cases of sudden accident, illness, faintness, drowning, or any other of the many unlooked-for ills and mishaps that 'flesh is heir to'—a species of knowledge that will improve the mind of the possessor, and may be productive of infinite good.

THE LAST OF THE STUARTS.

A MODERN ROMANCE.

III.—KING CHARLES III.

THERE were great preparations at Balquhalloch Castle for the worthy reception of the king; and but for the fact that the editor, who was also the sole reporter, of the *Aberdumble Warder* was at the time confined to his house from indisposition, the whole matter would no doubt have found its way into the papers. The Princess spared neither trouble nor expense. Two Aberdumble carpenters arrived, and erected opposite the porter's lodge a wooden triumphal arch. An Aberdumble painter followed after them, and inscribed upon the arch, in large red letters on a blue ground, a legend welcoming His Majesty to his own again. And when the painter had departed, the Princess, accompanied by Tom Checkstone and Father M'Fillan, drove into Aberdumble and hired all the flags in the town. She also subsidised the local brass band, the members of which she instructed to be at the castle at a given hour, prepared to play at her behest the stirring strains of *Who'll be King but Charlie*, and as many other Jacobite tunes as they knew or could learn meanwhile; and she further engaged four pipers, who were to stand just within the castle gate and salute her royal nephew in their most tremendous style. It must be confessed that the Princess was a trifle indiscreet. She undoubtedly laid herself open to a prosecution for treason-felony, if not indeed for treason of the highest and most deadly type; but fortunately for her, the Aberdumble people had grown accustomed to her eccentricities, and not a soul dreamt of gainsaying her will and pleasure in the matter. She therefore returned with a carriage-load of flags, which she caused to be festooned from the battlements. In her own boudoir and with her own fingers, she had long since worked in silk a faithful copy of the old royal standard of Scotland, and this she ordered to be run up on the flagstaff that surmounted the keep, whenever the king should set foot within the castle walls. Furthermore, she directed that at that auspicious moment her head-butler, assisted by one of her stable-boys, should begin

the firing of a royal salute from an old brass gun that stood upon the western wall; and in order to provide for this, she purchased in Aberdumble, Archie M'Pherson the ironmonger's entire stock of sporting-powder.

Betimes there came a telegram for the Princess. 'Shall be with you,' it ran, 'at noon to-morrow;' and it had been despatched by Charlie from the telegraph office in Fleet Street, London.

The Princess passed the night in a state of the utmost excitement. Instead of retiring to rest, she paced to and fro until daylight began to dawn; and it was only at Tom's urgent entreaty that she then consented to repair to her boudoir and lie down for a few hours. She had talked of attempting to raise the country side, and of going to the railway station at the head of her kilted and armed retainers, to welcome her nephew; but Father M'Fillan's more sensible counsel prevailed. He pointed out that nothing could be gained by undue haste, and that any ill-advised display of force would probably end in the speedy collapse of the movement long ere it could ripen and bear good fruit. The fact is that, but for the common-sense of the chaplain, the Princess would have done a thousand rash deeds. Fortunately, he had constituted himself her temporal as well as spiritual adviser; and being a man of extraordinary kindness and goodness, he had easily won the Princess's confidence. He regarded her as a harmless maniac, and believed that there would be no danger in humouring her within due bounds; nor did he conceal his opinions from Tom Checkstone, who, while professing to coincide with them, fully made up his mind to carry on the game until it should bring some decided advantage to himself and his friend Charles Stuart.

The Princess, accompanied by Father M'Fillan, and by Tom, who rode, drove to Aberdumble Station without the retinue of armed tenants; and, punctual to the appointed hour, the king's train arrived. The Princess would have respectfully kissed His Majesty's hand; but Charlie, with right royal condescension, embraced his aunt and kissed her upon both cheeks. Then, having graciously shaken hands with Tom and the priest, he entered the carriage, which was driven rapidly back to Balquhalloch. A few children followed it for a short distance, but they were soon left behind, and it was not until the castle was well in sight that any further signs of excitement and expectation became visible.

The first loyal greeting came from Daft Andy M'Gregor, an old fellow of eighty-seven. He had heard from his grandfather of the great doings of the '45; and fully believing that patriotism required it of him, he stood at his cottage door waving a rusty claymore, and flung his bonnet into the air as the carriage rolled quickly by. A more formal welcome awaited the king at the gateway of the castle. The porter's eldest son, in his Sunday clothes, walked gravely forward, bearing on a cushion the castle keys, and, with profound obeisance, presented them to Charlie, who as gravely received them and gave them into Tom's custody. At that moment the four pipers struck up; and the carriage, as it passed through the gateway, was saluted by a boom from the old brass gun. Tears stood in the Princess's eyes, as, giving her hand to Father M'Fillan, she alighted. 'Thank heaven for this day!' she exclaimed devoutly; while in a louder voice she added: 'God save the king!'

The cry was taken up by all the servants and tenants who had collected in the courtyard, and was accentuated by a second boom from the brass gun. Thus acclaimed, the king, supported by Tom, walked into the great hall, and was thence conducted by the Princess herself to his private apartments.

'The first thing,' she whispered to him, 'will be to hold a council.'

'Of course!' assented Charlie. 'We must hold a council at once. Let me see. May Tom—I mean Mr Checkstone—act as my private secretary until I can appoint some one else to the post?'

'Most certainly! Your Majesty's wishes are commands. I will send Mr Checkstone to you.' And the amiable Princess respectfully kissed her nephew and retired.

A minute later, Tom peeped in at Charlie's door and entered his room. The two young men at once burst into fits of suppressed laughter.

'It is really too bad!' exclaimed Charlie. 'You can't imagine how difficult it has been for me to keep my countenance.'

'Never mind; you have done very well so far,' returned Tom. 'I am to be your private secretary *pro tem.*, and you are to hold a council. I must therefore beg your Majesty to graciously nominate your councillors.'

'Whom shall I nominate? There is Father M'Fillan.'

‘And Alexander Gordon, the factor; he is a presentable sort of fellow.’

‘And my aunt, the Princess.’

‘Ah, the constitution doesn’t permit you to nominate women.’

‘Well, then, we four can hold the council, if the council must be held,’ said Charlie.—‘Now, how am I to comport myself?’

‘Father M’Fillan understands, of course, that we are only masquerading; but you mustn’t let him know that you have even the most distant designs on your aunt’s money-bags, for he would spoil the game in an instant if he did know it. As for Sandy Gordon—to him you must be the king, the whole king, and nothing but the king. If I were you, I should knight him. It would consolidate his loyalty.’

‘I’ll make him a baronet, if you like,’ assented Charlie. ‘It won’t cost any more.—But what are we to do in council?’

‘To decide upon your plan of action,’ said Tom. ‘And we may as well settle that here. The decision must be that no public steps are to be taken at present.’

‘Quite so.—But don’t, Tom, push matters too far. I only want to be able to marry Kate, you know; and really five hundred a year from my aunt would satisfy me.’

‘All right, old fellow; you shall have a thousand.—But now to business. I will go and announce the nomination of councillors. The council will meet in the drawing-room in half an hour. Shave yourself, and make yourself look as royal as possible. There was to be a valet to wait upon you, but he hasn’t yet arrived from Edinburgh.—How they are pounding away with that absurd brass gun! Fortunately, the powder has almost run out, so you won’t be troubled with your full salute.’ And Tom disappeared.

The first sitting of King Charles III.’s council was not a long one. The decision at which Charlie and Tom had already arrived was, of course, adopted. Sandy Gordon was offered, but modestly declined, the honour of knighthood; and Father M’Fillan was commissioned to inform the Princess that the king did not deem it expedient for the present to take any public steps to assert his authority beyond the loyal precincts of Balquhalloch.

The Princess, who for nights had been dreaming of rebellion, was grievously disappointed, and sought to bend her nephew's determination. After dinner, when for a short time she was alone with him, she talked the matter over.

'Well, my dear aunt,' said the king at last, 'I need not tell you that if we organise a rising, I must put myself at the head of it. And if I put myself at the head of it, I must of necessity risk my life. At present, as you know, I am not married. I am the last male of our house. Ought I not, before facing the danger, to look to the interests of my race, and indeed of the country? If I had a son, or even a daughter, I could go forth with a lighter heart to battle; for in that case my death would not mean the inevitable extinction of all our hopes. You, it is true, would succeed me, but only for a few short burdensome years; and with you the line would pass out of existence.'

'True,' assented the Princess; 'you certainly ought to marry, and to marry quickly. But whom can you marry? Who is worthy of you? Your royal grandmother was a Tudor, a worthy spouse for my father of sacred memory. But there are no Tudors now; and besides, you ought not to marry a cousin. I fear that you must seek an alliance on the continent, among the Bourbons or the Hapsburgs.'

'Why not among the English Plantagenets?' asked the king.

'Would that some remained,' ejaculated the Princess.

'But one does remain,' said Charlie. 'The fact is that I have had the good fortune to make the acquaintance of one whom I have determined to wed; and, my dear aunt, she is a Plantagenet. Mr Checkstone, who has the honour of her acquaintance, will tell you about her.'

'Indeed!' exclaimed the Princess. 'This news greatly rejoices me. Let Mr Checkstone write to her at once in my name and ask her to come and visit me. I will judge of her worthiness.'

'You had better write,' suggested the king. 'Why not write informally, as my aunt. You must remember that we cannot yet afford to risk anything by publicly hinting at my pretensions, much less by boldly proclaiming to the country and the powers that be that I intend to seize the crown.'

The Princess reflected; but she gave way, and, for once in her life, signed herself simply Henrietta Maria Stuart.

In due course Kate received the invitation. It was accompanied by a private letter from Charlie; and when she had read the two communications, she forfeited a month's salary and quitted her situation. A few days had to be spent in preparing for the visit to so large and grand an establishment as she knew Balquhalloch to be; but in less than a week after hearing from Miss Stuart, Catharine Plantagenet left London for Scotland.

IV.—THE COLLAPSE.

The Princess was charmed with Catharine Plantagenet, who, in truth, was as gentle and true-hearted a girl as could be met with anywhere; but when she became fully aware of the deception to which she was a party, it was with the greatest difficulty that Charlie persuaded her to refrain from telling her proud hostess the secret of her birth.

‘Well,’ said Catharine, ‘under any circumstances I won’t consent to take advantage of your aunt’s weakness. I hate false pretences. Your aunt ought to do something for you, I confess, but let her do it with her eyes open.’

Ere long, however, the Princess, upon her own initiative, made a proposition to which even Catharine saw no objection.

‘My dear,’ she said one morning, ‘I am getting old, and since you have been here with me, I have begun to feel that I should not like to be without you. Now I know perfectly well that Charles will be glad to stay here for the present; so, why don’t you make up your minds to marry and stay here together? When I die, the castle and everything belonging to it will be his. You need never regard yourselves, therefore, as trespassers upon my hospitality.’

‘And you really like to have me with you?’ asked Catharine.

‘Certainly, my dear.’

‘For my own sake, I mean?’ added Catharine.

‘Yes, for your own sake, and quite apart even from the fact that Charles loves you. I shall speak to him about it.’ And speak to him she did.

‘You will be quieter and better off here than in London,’ she said; ‘and you will be able to mature your plans for the future. You and Catharine shall

have a separate establishment for yourselves; there is plenty of room for all of us. And if you have any hesitation on the score of money matters—which, after all, trouble the highest as well as the lowest—I may set your mind at ease, my dear Charles, by telling you that I have determined to give Catharine on her wedding-day a hundred thousand pounds by way of jointure. When I die, the rest will be yours.’

‘You are very good, aunt,’ exclaimed the king, who was fairly overcome by his relative’s liberality. ‘Yes; nowhere can we be happier than here. But let us be married quietly.’

‘By all means! Father M’Fillan shall perform the ceremony in the chapel. Get Catharine to name a day—the sooner the better.’

Charlie talked over the matter that very evening with his sweetheart, and an arrangement was soon come to between them. The wedding was fixed for an early date; a few favoured guests were invited; and in due course Charlie and Catharine became man and wife, Tom acting as best-man, and Sandy Gordon, who, on account of his age and patriarchal beard, seemed to be peculiarly fitted for the part, giving away the bride. There was, of course, a feast for the tenantry; and the brass gun on the wall was again fired—this time until it burst; but, as the Princess regretfully said, the ceremony was not worthy of the event. It ought to have taken place at Holyrood or Westminster Abbey.

Charlie and Catharine went to Edinburgh for their honeymoon; and when they returned to Balquhalloch, the castle settled down once more into its normal condition of peace and quietness. Tom and the Princess spent much of their time in the library, working hard at the family history; and the young couple, with nothing to worry them, and only themselves to think about, passed a delightful existence, which seemed as if it could never become wearisome.

But matters could not go on for ever in this way. The Princess in time began to ask Charlie about his plans. ‘Will arms be required?’ she wanted to know. ‘Will there be uniforms for the troops? What hope is there of foreign assistance? Can the officers and men throughout the country be bribed?’ And above all, ‘When are you going to rise and strike for your rights?’ In short, the situation threatened to become critical. And when, nearly a year after his wedding, Charlie found himself the proud father of a boy, he

realised that he must either act, or permit his good aunt to scorn him as a weak-kneed, cowardly shadow of a king.

The auspicious event caused the Princess to be unusually active. She desired, ere the grand stroke should be dealt, to be in a position to publish abroad a full and complete pedigree, tracing the descent of the Stuarts of Balquhalloch from the royal Stuarts of Scotland; and hearing that a number of old records bearing upon the question were for sale in London, she despatched Tom Checkstone thither with *carte blanche* to buy whatever he could lay his hands upon. Tom was absent for ten days; and when he reappeared, he had with him a large chest full of dusty, mouldy, discoloured documents. These treasures were conveyed to the library, and for a week the Princess almost lived among them.

One day Charlie and Catharine, who was now convalescent, were sitting at luncheon, when, without warning, the Princess burst into the room. She was very violently excited. Her gray hair had fallen loose, her cheeks were pallid, and her hands were clenched convulsively.

‘What is the matter?’ exclaimed Charlie and Catharine, both rising together and rushing to support their aunt.

‘Matter!’ she cried—‘matter?’ and she began to weep hysterically.

‘Tell me,’ implored Catharine. ‘What can we do?’

But Tom, who had followed the Princess, and who now appeared in the open doorway, soon explained the cause of the outbreak.

‘Look here!’ he said, as he held out a yellow parchment. ‘It is a terrible blow to your aunt, Charlie. There has been some mistake. You are not descended from the royal Stuarts at all. A similarity of names and some careless copying are responsible for the error.’

Charlie seized the manuscript, and having hastily glanced at it, threw it aside, and went to his aunt, who was already being attended to by Catharine.

The Princess had fainted; but ere long she recovered, and was able to tell her version of the story. She had been completing the pedigree; she had almost arrived at the last link, when the whole chain had been snapped by

this hideous discovery. She would never get over the shock. To think that after all she was a nobody! It was too dreadful!

They led her to her own room, and in time succeeded in calming her. Then, in order to convince himself, Charlie carefully examined the parchment. Its statements could not be gainsaid. The Stuarts of Balquhalloch had no connection with royalty; and he would not now be required to seize the throne of Great Britain. To him the revelation came, it must be remarked, as a welcome relief; but for days and weeks it made his poor aunt miserable; and when she finally reconciled herself to her lot, it seemed as if her energy and pleasure in life had departed for ever. Indeed, she never entirely got over the blow, and at the beginning of this year she died.

Charlie and Catharine were with her to the last, and she bequeathed everything to them. Balquhalloch, therefore, is now theirs; and Tom Checkstone, who, rightly or wrongly, regarded himself as Charlie's good genius, holds sway as his friend's secretary, man of business, chum, and general factotum.

THE MONTH: SCIENCE AND ARTS.

ON the 27th of August, the British Association commenced its fifty-fourth annual meeting—not this year, however, on British soil, but at Montreal in Canada. Some hundreds of members travelled from this country to be present at the meeting. Both on the part of the city and of the Dominion, the reception of the Association has been everything that could be desired by its members. Montreal itself raised forty thousand dollars towards defraying the expenses of the visit, and three hundred of the members were, besides, received as guests into private houses. The new President, Lord Rayleigh, Professor of Experimental Physics in the University of Cambridge, delivered the opening address, in which he sketched the progress that had been made in certain important branches of science. The final meeting of the Association was held on the afternoon of September 3, and was largely attended, about two thousand persons being present. Lord Rayleigh in addressing the assembly, said that no meeting had been held in which the Association had been provided with such spacious rooms. Resolutions in favour of the erection of a free public library in Montreal, as a memorial of the visit, were then passed, and a large amount of money was immediately promised in aid of this object, among the donations being one of ten thousand pounds. The total money grants in favour of scientific investigations made at the Montreal meeting were fifteen hundred and fifty pounds. The tickets issued for members of the Association for this session numbered seventeen hundred and thirty, and the money received amounted to eighteen hundred pounds.

Within the last twenty-five years, and more especially since the Franco-German war, when the French made such good use of balloons, there have been somewhat frequent rumours that the problem of aërial navigation, comprised in the possibility of guiding and propelling a balloon in a given direction, had been solved. The machine in each case is carefully described, and generally it is represented as having risen gracefully in the air, travelled about a little, and then returned to its starting-point. Then, nothing more is

heard of it. Such an event is said to have occurred last month in France. The gas-vessel—it can hardly be called a balloon, for it is cigar-shaped—is nearly two hundred feet long. A platform is hung below, upon which is a screw propeller, worked by a dynamo-machine and a large rudder. This description tallies almost exactly with the form of so-called steerable balloons which have been constructed, tried, and found useless by M. Giffard, M. Tissandier, and others in previous years. The French government have spent much money in experimental ballooning, and this last achievement is the result. Perhaps the authorities were obliged to show something for the money that was being spent, but we fear that that something is not anything new or profitable. Until an aërial machine be produced which shall make its way *against* strong currents, balloon navigation will remain as it has hitherto been.

Here is a clever American notion, and one which will probably have a wide application. It consists of a noiseless door-closer. In the ordinary metal or india-rubber spring, so commonly fixed to doors, the greatest energy is exerted at first, and the door generally slams with a noise which is very distressing to any one with nerves. In the new arrangement, the spring is fixed to the piston attached to a small air-cylinder, so that as the door closes, the resistance of the air in the cylinder checks its motion before the terrible bang arrives. A small opening in the cylinder then lets in the air, so that the spring once more asserts its authority with sufficient persuasion to gently close the door.

After the terrors excited by the alleged danger of using arsenical wall-papers, it is rather a relief to read the opinion of Mr R. Galloway, who has written an article upon the subject in the *Journal of Science*. ‘Has it,’ he asks, ‘ever been proved that persons who inhabit rooms the wall-paper of which is stained with emerald green, suffer from arsenical poisoning?’ He then points out that the injurious effects, if any, must be due to the mechanical detachment of the pigment from the paper, and that such homœopathic doses of the substance as could be carried by the air, would be totally different from the effects which arise from larger doses of arsenic. Moreover, he has made inquiry as to any cases of poisoning occurring during the packing of this finely divided pigment—during which operation the packers are surrounded by clouds of its dust—and could hear of none. Mr Mattieu Williams, a well-known writer on Science, is also of opinion

that 'arsenical wall-papers' are practically harmless. We are glad to record these opinions, for the tendency of the present time is to point out lurking dangers in every direction, until one is apt to wonder how our forefathers, in their happy ignorance of sanitation, ever contrived to reach adult age.

At Reading, this autumn, a honey-fair is to be held, when prizes will be distributed to beekeepers who work on humane and advanced principles, and also to those who can show the greatest amount of unadulterated honey raised in a Berkshire hive. Such a show as this is worthy of every encouragement, for honey fetches a high price, and so does wax, even in these days of cheap sugar and composite candles. It thus becomes possible for the intelligent cottager to add considerably to his scanty means; and if he can be taught that honey can be won without periodical destruction of bees and comb, so much the better. There is some complaint that the newfangled hives, efficient though they be, are too expensive to supersede the old straw skep. The British Beekeepers' Association might well turn their attention to this aspect of the matter.

Last year, Professor Huxley stated it as his opinion that no act of man could possibly influence the increase or decrease in the number of sea-fish. This was in answer to the gloomy anticipations of many that the herring and other fisheries would be gradually annihilated unless our fishermen were compelled by law to observe certain conditions. So far as herrings are concerned, the recent enormous catches have shown that there are fish as good and plentiful in the sea as ever yet came out. Last month, we saw these fish in splendid condition being sold at the Farringdon Fish Market, London, at one penny per dozen. By the way, can any one explain why, in these days of refrigerators and cheap ice, eighty-six tons of fish should be allowed in one month to become—at Billingsgate—unfit for human food?

At a flower-show at Frome the other day, prizes were offered by Miss Ormerod, the consulting entomologist to the Royal Agricultural Society of England, for the best collection of food-plants injured by insects, accompanied by samples of the injurious creatures themselves, and a short written account of the nature of their depredations and the preventive measures to be adopted in dealing with them. There was only one competitor, Mr Herbert Haley of Frome; but the collection which he showed was a very complete one, and was most highly commended by Miss Ormerod. As this was the first injurious-insect competition in this country,

and was probably known to comparatively few persons, we need not be surprised at the want of competitors. Ten years ago in Paris, a similar Exhibition took place, in which nearly four hundred competitors took part. The exhibits included useful as well as injurious insects, which were divided into separate classes. Such competitions ought to be productive of a great amount of good.

Recent experiments have led to the adoption of many alterations in the torpedo system, which is likely to play such an important part in naval operations of the future. Hitherto, the torpedo—a huge fish-like case to hold explosives, and containing within itself an air-engine for propelling it through the water—was thrust from a ship's side below the water-line. But it was found in practice that it was impossible to fire the agent of destruction in a straight line, especially if its mother-ship were under way. In the new method, the torpedo, which is sixteen feet long and fourteen inches in diameter, is fitted into a steel tube just large enough to contain the projectile. The pressure of a key admits highly compressed air to this tube, and the torpedo is shot out of an open port on exactly the same principle that a pea is projected from a pea-shooter. But in the case of the torpedo, its little but powerful engine is set to work the moment it reaches the water, and away speeds the torpedo on its terrible errand.

Professor Tuck of New York has constructed an electric torpedo boat, which will render submarine warfare very terrible, if the hopes raised by its recent trial are fulfilled. It is made of iron, in the shape of a cigar, or rather pointed at each end, and is thirty feet in length. It can travel on the surface of the water, or several feet below, at the will of its commander. The torpedoes are carried outside the vessel, and can be detached by the action of an electromagnet, when it is desired to let one rise to the surface against any ship that may be lying overhead. By means of attached wires, the torpedo can be exploded when the submarine boat has retired to a position of safety. Jules Verne's clever romance *Thirty Thousand Leagues under the Sea*, seems here to come into play.

Under the supervision of Mr Preece, the well-known electrician to the Post-office, an experiment of great interest has been recently conducted at Wimbledon, near London. The object of the experiment was to ascertain the best method of lighting streets by electricity, the lamps employed being the incandescent pattern advocated by Edison, Swan, and others. Some of the

lamps were placed singly, others on poles twenty feet high, while at the same time the efficiency of different kinds of reflectors was tested. The cost is calculated to amount to one farthing per lamp per hour for each unit of light, valued at ten candles. Now, the cost of gas for a similar amount of light is one-fifth of a penny, so that the difference in expense between the two systems is not very great. On the other hand, the advantages of the electric light in the open air, where no question of impure products of combustion need come in, almost disappear; gas, therefore, still fulfils the required conditions.

It is to be hoped that the new regulations for the prevention of collisions at sea, which have just been published, will have the effect of reducing those calamities, which have of late become fearfully familiar. The twenty-seven articles contained in these regulations refer to lights, sound-signals for fog, steering and sailing rules, precautions to be adopted, and special rules for squadrons and convoys. We may call particular attention to article nineteen, which indicates how one vessel can signal to another by a steam-whistle. Thus—one short blast to mean, ‘I am directing my course to starboard;’ two short blasts, ‘I am going to port;’ three to mean, ‘I am going full speed astern.’ Any one will readily remark how such signals could be added to almost indefinitely. Indeed, it is simply the method adopted in the army for flash-signalling with the heliograph, adapted to sound-signalling with the steam-whistle. In such a system, unfortunately, there are few, in moments of danger, who can keep their heads cool enough to avoid making perilous mistakes. It is somewhat like talking quietly when the house is on fire.

About four years ago, the startling scheme of carrying ships upon a specially constructed railway track bodily across that little neck of land which ties together the two Americas was conceived by Mr Eades. This scheme was communicated to the British Association at York in 1881, and although the idea seems a novel one, like most engineering enterprises it can be doubtless accomplished if investors can be made to believe in its power of paying a good dividend. The suggested route would require a track one hundred and thirty-four miles in length, which must be laid with a compound railway of extremely solid construction. The worked-out details of the method of raising a ship on a pontoon, transferring it to a travelling cradle, and finally committing it to the deep once more, after its journey on

dry land, are most ingenious. Mr Eades, who hails from the United States, is now in England, endeavouring to interest capitalists in his proposals.

The demolition of the old law-courts which adjoined Westminster Hall, has brought into view one of the most interesting pieces of stonework belonging to Norman times that can be found in London. The state of preservation of the wall of the old hall, upon which the marks of the mason's tools are still visible, is due to the circumstance, that from a very early time it was under cover, for a cloister extended along the whole length of the building on this side. Mr Pearson, the architect, was lately requested to report upon the subject, and to suggest the best method of restoration compatible with the preservation of this unique relic of the reign of William Rufus; and it has been determined to restore the cloister as it originally stood. According to the opinion of Mr Shaw-Lefevre, the First Commissioner of Works, the edifice when completed will, with the Houses of Parliament and the old abbey adjacent, form one of the grandest groups of buildings in Europe.

The attractions of South Kensington Museum have lately been added to by the opening of a room containing a collection of antique casts, which have been collected and arranged by Mr W. C. Perry. This collection numbers about three hundred specimens, which illustrate the whole historical range of ancient art. Such a record of the plastic art of ancient times is of deep interest to the archæologist, as well as of immense value to the art student. The arrangement of the specimens is mainly chronological, and where one or two casts are, on account of inconvenient size, not shown in their right place, it is in consequence of want of space. We may venture to hope that at some not distant date, better accommodation will be found for this valuable and interesting collection.

The Great Western Railway has always been famed for the wonderful engineering difficulties which were grappled with by the daring Brunel, and many evidences of his skill are apparent to the traveller on that line. But even Brunel did not conceive the bold idea of piercing a tunnel twenty-six feet in diameter, and four and a half miles in length, beneath the bed of the Severn. But this great work has now been in progress for some years, and the operations latterly have been pushed forward with such rapidity by three thousand busy men, that its completion may soon be looked forward to. The tunnel will shorten considerably the distance between London and South Wales. It is constructed so as to dip considerably towards the centre, to

which point any water will naturally gravitate. Here it will enter a drainage subway, which will carry it to the Welsh side, to be pumped up into the river. The great difficulty which the workers have had to contend with is the irruption of vast bodies of water from local springs. The crown of the tunnel lies at a depth of from eighty to a hundred feet beneath the bed of the river.

In these days of quick communication by telegraph and telephone, it is strange to see how it becomes occasionally convenient to employ 'the bird of the air' to 'carry the voice.' In Haddingtonshire, at the Penston Colliery, messages are carried from the pits to the offices, a distance of more than six miles, by pigeons, and they accomplish the work in about as many minutes. Telegrams are found to take about an hour in executing the same business, and telephones are inadmissible, because as yet no plan has been found by which the sounds can be permanently recorded.

In a recent lecture on Cholera and its Prevention, Professor de Chamont called attention to the very common and erroneous idea that tobacco-smoke, camphor, and other strongly smelling compounds act as disinfectants. He pointed out that although chlorine, sulphurous acid gas, and carbolic acid may under certain conditions be safely regarded as true disinfectants, the best and most efficient known is *fire*. He also, in speaking of sulphurous acid gas, generated by burning sulphur, showed that a ready way of facilitating combustion was first of all to pour upon the brimstone a little alcohol.

Mr Graham, who recently gave an account of his mountaineering experiences in the Himalaya, seems to have negatived some of our preconceived notions regarding the difficulty of breathing at high altitudes. At an elevation of more than four miles above the sea-level, Mr Graham and his companions felt no inconvenience in breathing except what might be expected from the muscular exertion they had gone through. Loss of sight, nausea, bleeding at the nose or ears, and other unpleasant symptoms often described by travellers, were entirely absent. But the heart was sensibly affected, its rapid pace being easily perceptible, and its beatings quite audible. It may be remembered that Mr Glaisher and Mr Coxwell, in the course of an experimental balloon ascent some years ago, nearly lost their lives by the effect upon their breathing organs of the highly attenuated atmosphere to which they had risen. But the altitude then reached was about double that attained by Mr Graham in the Himalaya.

A scheme has been proposed for the construction of an Indo-European railway, the chief novelty of which is the adoption of a route along the south shore of the Mediterranean. The line would utilise the railroads of France and Spain. Then there would be steam-transit from Gibraltar Bay to Ceuta in Morocco. Here would be the terminus proper of the international railway, which would be in connection with the lines already laid in Algeria and Tunis. The route would be continued through Tripoli to join the Egyptian lines, and eventually along the coast of the Persian Gulf to Kurrachee in India. Here, of course, contact would be made with the great Indian railway system. Preliminary surveys have been already made, and the nominal capital of the undertaking is fixed at ten millions sterling.

At the late meeting of the British Association in Canada, a very curious contribution to our knowledge of carnivorous plants was made by Professor Moseley, as a result of certain experiments he has made with the water-weed *Utricularia vulgaris*. This plant is furnished with small pear-shaped bladders, which at certain seasons are charged with air, and cause the weed to rise to the surface of the water. This movement has hitherto been supposed to be connected with the phenomenon of fertilisation. But Professor Moseley points out that each bladder has an opening closed by an elastic door, which will easily yield to the pressure of a small fish; and that any unfortunate intruder is either caught bodily, or can be securely held a prisoner by head or tail until dead. That there is here anything analogous to digestion as seen in other carnivorous plants, such as the *Dionæa*, &c., does not appear; but it is thought probable that the decomposing animal matter may contribute eventually to the life of the weed.

An invention of considerable importance in connection with the probability of saving life at sea has made its appearance during the month. This consists of an adaptation of the use of oil at sea to the ordinary life-buoy. Round the inside of the buoy is a brass reservoir filled with oil. This is so arranged that when the buoy is hung upon the vessel's side no oil can escape; while the moment it assumes a horizontal position, as, for instance, when it is thrown into the sea, the oil flows freely, and the water all around the buoy is rapidly covered with a thin film. This soon widens into a large circle, within which, of course, the waves are unbroken, which enables persons to be the more easily secured by the ship's boats. Since it is a well-known fact that in rough weather, when the cry 'Man overboard!' is oftenest heard, life-buoys

are frequently useless, as even the strongest men are commonly washed off them, this practical adaptation of the use of oil at sea will probably prove of signal importance. It could, we imagine, also be readily applied to many of those improvised sea-rafts and similar appliances, and render them of great value in rough weather. It was the fault of many of those ingenious contrivances of this kind which were to be seen at the Fisheries Exhibition last year, that no one could possibly live on them in broken water, and this objection the use of oil in this way would certainly obviate. It should be noticed, however, that the chief value of the invention consists in the arrangement for the oil to flow automatically.

In addition to the electrically lighted colliery in South Wales, noticed in last 'Month,' we hear of another in Lanarkshire belonging to Mr John Watson, Earnock, near Hamilton. There may possibly be other workings thus illuminated throughout this country; and there is no doubt that ere long the brilliant and comparatively safe electric light will be generally adopted underground.

In Prussia also, as we learn from a contemporary, the electric light at the Mechernich Mines has now had a fair trial for more than three years, and has proved a complete success. The expectation that it would both facilitate the operations and increase their security, has fully been realised, and an extension of the plant is now being carried out. An open working two thousand feet long, one thousand feet wide, and over three hundred feet deep, in which three hundred men and twenty horses are continually occupied, was first to be supplied with the electric light, and it was a question whether arc lamps would answer for this purpose in the smoky atmosphere caused by blasting operations. For the first experiments, arc lamps of three thousand and one thousand candles were used, with the positive carbon in the lower holder. The effect was brilliant, yet the light did not penetrate the white smoke cloud which collects at the upper wall immediately after the shot. But as the smoke settles within ten minutes, it was thought advisable to acquiesce in this interruption of a few minutes, and to use smaller lamps of three hundred and fifty candles, which proved quite efficient. Of these, there are ten in use, with about ten thousand feet of lead cable, the cable being partially elastic, as the lamps with their wires have to be removed when the blasting is to take place. The lamps were originally supplied with hexagonal lanterns with obscured glass to protect

the eyes of the miners. The glasses were, of course, soon broken, but no complaints are said to have been made about the naked electric lights.

The speech-recorder would appear to be an instrument of no small importance, if it is able to do in a practical manner that which the title of a patent recently applied for by Mr W. E. Irish would lead us to suppose. The title of the patent is as follows: 'A system or method and means of receiving and recording articulated speech and other sounds transmitted telegraphically, telephonically, or otherwise, by the aid of electricity.' The transmitting, as by telephone, and recording of speech in characters which may be easily read, would be of incalculable value. If this instrument fulfils what is claimed for it, the anticipations once hoped for in the phonograph will be realised, and in the future we may expect to see business-men talking their correspondence into a box in which mechanism, by the aid of electricity, records the same on paper, which may be forwarded as a letter. Moreover, literary men will be saved the drudgery of the pen, and have their thoughts recorded as rapidly as they can convey them to the instrument. The system of natural phonetic signs, which we should expect this instrument to describe, may also be the means of influencing spelling and of simplifying the phonographic difficulties of the language. Applications innumerable suggest themselves to us to which such an apparatus may be applied; the verifying and duplicating of orders received and sent telephonically, would form no small item in the advantages to be derived from such a system.

According to the *Journal de Rouen*, quoting from the *Polytechnische Zeitung*, the recent invention of M. Verk, by which is produced the effect of any metal on felt, is likely to become extremely useful when applied to theatrical stage properties, as, besides being inexpensive, the articles so treated are not materially increased in weight. The things intended to assume a metallic appearance are first of all covered with a layer of felt, which is coated over with a resinous substance mixed with plumbago or blacklead. This is left to dry, and is then passed over with a hot iron. The article is next rubbed with pumice-stone, which produces the effect of burnished steel. If copper, bronze, or silver is wished to be imitated, the felt—which is rendered a conductor by its coating—is covered readily by immersion in a galvano-plastic bath.

OCCASIONAL NOTES.

TELEGRAPHING EXTRAORDINARY.

A CONTEMPORARY supplies some interesting particulars as to the number of words transmitted by telegraph to all parts of the kingdom on the occasion of the Prime Minister's recent visit to Edinburgh. On the evening of Mr Gladstone's arrival, press messages containing over seventeen thousand words were handed in at the telegraph department of the General Post-office; but the actual number of words transmitted was over sixty-seven thousand, owing to the fact of the same report being sent to more than one newspaper. Mr Gladstone's visit to the Forth Bridge works led to the transmission of twelve thousand words, and his movements on the following day to nineteen thousand. On the occasion of his first speech on Saturday evening (August 30) in the Corn Exchange, sixty-two thousand four hundred and seventy-one words were handed in, and one hundred and thirty-eight thousand four hundred and forty-five transmitted. The number would have been greater had not Sunday intervened, allowing of the transmission of many messages by train. On Monday evening (September 1) the press messages reached the enormous number of one hundred and seventeen thousand words, causing the transmission of about four hundred and twenty-seven thousand words, the largest number ever transmitted on any one night from Edinburgh. After the Waverley Market speech of Tuesday night (September 2), one hundred and seventy-two thousand eight hundred and twenty-one words were transmitted.

On Monday evening, when the strain was heaviest, one hundred and thirty operators were at work, and in spite of the constant stream of messages the department kept abreast of the reporters. As many as four towns in the same telegraphic circuit were enabled to read almost the same message at the same time. The message having been 'punched' on long slips of prepared paper, the plan was adopted, instead of running it entirely through one machine, of taking the slip out of the first machine after it was three yards clear, and running it into a second and a third.

CASUALTIES ON THE BRITISH COAST.

A blue-book on sea-casualties to British vessels from July 1, 1882 to June 30, 1883, contains the Report of the Marine Department of the Board of Trade, showing that the total number of vessels belonging to the United Kingdom to which casualties occurred (total losses and serious and minor casualties) was 5409. This is higher by one hundred and eleven than in 1881-82, when it was 5298, and higher than any year since 1876-77, when it was 5801. The number of total losses was eight hundred and ten (tonnage 277,490). This is lower than the previous two years, but higher than in either of the four years preceding. The decrease in the last two years is in sailing-vessels (classed and unclassed) and in unclassed steamships. There is, however, a large increase (twenty-five per cent.) in the total losses of classed steamships. The number of serious casualties not amounting to total losses was 1268, and was lower than any of the previous six years, except 1879-80, when it was slightly higher. The decrease is entirely confined to sailing-vessels. The number of steamships (five hundred and seventy-one) to which serious casualties occurred is larger than in any of the previous six years, and the tonnage of the vessels affected is nearly one hundred and fifty thousand more than in 1876-77. From 1877-78 there is a steady annual rise in serious casualties to steamships from four hundred and six in 1877-78 to five hundred and seventy-one in 1882-83. In 1876-77 they were four hundred and ninety-three. The loss of life in vessels belonging to the United Kingdom was 2501 in 1882-83, or seven hundred and seventy-six less than in 1881-82, but was more than the loss in each of the five years preceding 1881-82, and was three hundred and thirty-one more than the average for the six years. Of these 2501 lives, 1463 were lost in missing vessels. The number of missing ships was one hundred and fifty-two, namely: Sailing-ships, 133; tonnage, 32,995; lives lost, 1080: steamships, 19; tonnage, 14,626; lives lost, 383—total lives lost, 1463.

‘YES.’

A LITTLE rain,
The sun again,
 A shadow;
A summer day,
Some new-mown hay,
 A meadow.

A girlish face,
A matchless grace,
 And beauty;
We spend the day
In making hay—
 Sweet duty.

Some fading flowers,
Some happy hours,
 But fleeting.
A week of rain,
And then again
 A meeting.

One quick shy look,
A rippling brook,
 Some clover;
A sky of gold,
The story old,
 A lover.

A fair sweet maid,
A short word said;
 What is it?
I try my fate,
And not too late

To miss it.

The years have gone,
And still loves on
That lover;
He loves always,
As days and days
Pass over.

A loving wife,
A long, long life
Together,
Have made him bless
That shy sweet 'Yes'
For ever.

NORA C. USHER.

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